

1M sample and assume that efficiency gains for any task in our sample, even those with just 15 observations out of one million, the implied aggregate labor productivity growth over the next decade would be roughly 5% percentage points per year—a mechanical increase based on a the much larger set of tasks included.

⁷ As before, this result is based on applying [Hulten's Theorem](#) to task-level productivity shocks and assuming that the corresponding one-time increase in total factor productivity materializes over the course of a decade alongside capital deepening effects.

⁸ As a reminder, for aggregating to implied labor productivity we calculate task-level efficiency gains as the log difference between human time without AI and with AI. There are certainly other ways to adjust based on task reliability. If tasks in our sample are composed of sub-tasks with heterogeneous AI applicability, and workers optimally deploy AI only on sub-tasks where it is effective, then scaling the efficiency gain by the success rate captures the extensive margin of AI adoption within a task.

⁹ We use a CES (constant elasticity of substitution) production function to aggregate task-level time savings to economy-wide productivity impacts. The elasticity parameter σ governs how easily workers can substitute between tasks. When $\sigma=1$, we apply Hulten's theorem directly: the aggregate productivity gain equals the wage-share-weighted sum of log speedups across tasks. For $\sigma \neq 1$, we use a two-level aggregation: first, within each occupation, we compute an occupation-level speedup as a CES aggregate of task speedups weighted by time fractions, using $\rho=(\sigma-1)/\sigma$. Then we apply Hulten's theorem to these occupation-level speedups. When $\sigma < 1$ (complements), productivity gains are bottlenecked by tasks with the smallest speedups. When $\sigma > 1$ (substitutes), workers can specialize in tasks where AI provides the largest speedups, amplifying aggregate gains. For tasks without observed AI speedup data, we assume no productivity change. We thank Pascual Restrepo for suggesting this particular exercise.

Concluding Remarks

This fourth Anthropic Economic Index report introduces economic primitives—foundational characteristics of AI use—that show how Claude is used by both consumers and firms. We use Claude to estimate the extent to which usage varies along these dimensions; these measures are directionally accurate and, taken together, provide important signals even if individual classifications are imperfect.

Our findings carry significant implications for how AI will reshape economies and labor markets. Notably, Claude tends to be used more, and appears to provide greater productivity boosts, on tasks that require higher education. If these tasks shrink for US workers, the net effect could be to deskill jobs. But these impacts depend crucially on complementarity across tasks, and whether increased productivity at a certain task may increase the demand for it.

At the global level, the strong relationship between per capita income and usage patterns—with higher-income nations using Claude collaboratively while lower-income countries focus on coursework and specific applications—suggests that AI’s impact will be mediated by existing institutional structures rather than unfolding uniformly. Geographic diffusion patterns reinforce this picture. Within the US, per capita usage has converged slightly; globally, diffusion is slower. Combined with income-driven differences in how AI is used, this raises questions about whether AI will narrow or widen international economic gaps.

Equally important to the patterns documented here are potential changes across this and subsequent reports. As AI capabilities advance, Claude’s success rate may increase, usage patterns may show greater autonomy, users may tackle new and more complex tasks, and tasks that prove automatable may graduate from interactive chat to API deployment. We will track these dynamics over time, providing a longitudinal view of AI’s role in the economy.

Building on prior releases, this edition significantly expands both the scope and transparency of usage data we share, including task-level classifications along new dimensions and regional breakdowns globally for the first time. We publish this data to enable researchers, journalists, and the public to investigate novel questions about AI’s economic impacts that can form the empirical foundation for policy responses.

How willing users are to experiment with AI, and whether policymakers create a regulatory context that advances both safety and innovation, will shape how AI transforms economies. For AI to benefit users globally, expanding access alone will not suffice—developing the human capital that enables effective use, particularly in lower-income economies, is essential.